

On the Use of the ubcdiss Template

by

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The following individuals certify that they have read, and recommend to the Faculty of Graduate and Post-doctoral Studies for acceptance, the thesis entitled:

On the Use of the ubcdiss Template

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Abstract

1. Shifted phenology with ACC extends GS; expected growth increase
2. Recent studies failed to find this relationship; with drought and competition restraining growth
3. Objective: understand if trees grow more with longer GS where drought and competition were limited
4. Leveraging 9 years of phenology from 2 studies, both in an arboretum and by relating it to tree growth
5. Juveniles grew more; adults did not change their growth with longer GS
6. Our results reveal a more complex growth response to climate change than currently assumed
7. Not massive responses and when they do, it's more for juveniles than mature: good for forest regeneration, potentially bad for forest carbon storage.

Lay Summary

1. Climate change lengthens the time during which trees can grow
2. This should increase their growth; but evidence shows this may not be the case
3. Here we use young and mature broadleaf trees to understand whether they grow more with longer growing seasons
4. Young trees grew more, and appear to respond more to end of season temperature than early season
5. Mature trees did not grow more, which suggest that they are less responsive than their young counterparts
6. This highlights that climate change may not harm forest regeneration, but could limit future carbon storage as mature trees store most of forest carbon.

Preface

1. Chapter 1

- (a) I leveraged phenology data collected by Dan Buonioto and Cat Chamberlain in Boston, a project started by Dr. EM Wolkovich
- (b) The research questions and project were proposed by EMW
- (c) I processed, scanned, measured the tree cross-section and cores previously collected by Neil Pederson, Dan, EM Wolkovich and Deirdre Loughnan
- (d) Victor VdM and EM Wolkovich helped me with building the model
- (e) I performed the analyses and created all the associated figures
- (f) The following co-authors reviewed the manuscript: DL, VVDM, DB, CH, NP and EM Wolkovich.
- (g) No generative AI was used to write this thesis. I used GenAi for performing simple repetitive coding tasks and for help with the figures, data cleaning and improve coding workflow efficiency.
- (h) Should be submitted for publication by the time this gets sent to GPS

2. Chapter 2

- (a) I leveraged data collected from citizen scientists and cored the trees with XW
- (b) Project was conceptualized by EMW and me
- (c) I collected, processed, scanned and measured all the cores and performed the analyses
- (d) The following co-authors reviewed the manuscript: DL, VVDM, DB, CH, NP and EM Wolkovich.
- (e) GEN AI for this manuscript same as Chapt 1
- (f) This work is intended for publication

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Glossary

This glossary uses the handy `acroynym` package to automatically maintain the glossary. It uses the package's `printonlyused` option to include only those acronyms explicitly referenced in the $\LaTeX$ source. To change how the acronyms are rendered, change the `\acsfont` definition in `diss.tex`.

Acknowledgments

Thank those people who helped you.

Don't forget your parents or loved ones.

You may wish to acknowledge your funding sources.

Chapter 1

Wildchrokie

Introduction

1. Phenological shifts with climate change
 - (a) Shifts in spring/autumn modify start and end of season (Cleland *et al.*, 2007; Lieth *et al.*, 1974; Menzel *et al.*, 2006; Parmesan & Yohe, 2003; Woolway *et al.*, 2021)
 - (b) Early spring + delayed autumn should increase growth
2. Positive relationship between season length and growth is important
 - (a) Carbon cycle models rely on this positive relationship
 - (b) This could impact climate solutions
3. Maybe longer GS don't increase growth
 - (a) Dow showed longer GS didn't increase growth locally
 - (b) Cabon showed decoupling between GPP/Growth
4. Longer GS does not necessarily mean more favorable conditions for growth throughout the season
 - (a) Longer seasons are not necessarily warmer
 - (b) Earlier SOS, also means earlier EOS—gains cool early season days (little increase in GDD) but miss warmer late season days (Figure 1.1)
 - (c) Water availability may constrain growth
 - (d) Competition may offset extra days to grow especially for below canopy trees
 - (e) Length and speed of growth is physiologically (or genetically?) controlled
 - (f) And how much phenology can track these changing conditions matters
5. Leaf and wood phenology are not symmetrically plastic

- (a) Leaf phenology in the spring is very plastic (well studied)
 - (b) Leaf phenology and growth are synchronized = good proxy for growth SOS and EOS
 - (c) However, this does not say anything how much trees grow in a year
 - (d) How flexible growth is may change between species, but also across populations
6. Local adaptation to inform growth with climate change
- (a) Trees from northern provenance may grow less with longer seasons
 - (b) Budset is strongly locally adapted
 - (c) Common gardens are important for climate change and tree growth
7. Tree growth will vary across species but will also depend on future environmental conditions
- (a) To understand GS and growth relationship, need to look more precisely at when trees grow during the GS
 - (b) If growth is not plastic enough, trees may not be able to track favorable changing conditions in the season
 - (c) Varies across species and population, with the latter having a potential to inform how species will respond to future conditions
 - (d) Juvenile trees store little carbon, but are crucial for forest regeneration capacity
8. Answer how GS and growth relate using juvenile trees in a common garden
- (a) Disentangle GS length vs GS thermal conditions
 - (b) Growth with tree rings, SOS and EOS with leaf phenology
 - (c) Calendar season: days between SOS and EOS and thermal season: GDD between these days
 - (d) Will look at different species from different provenances

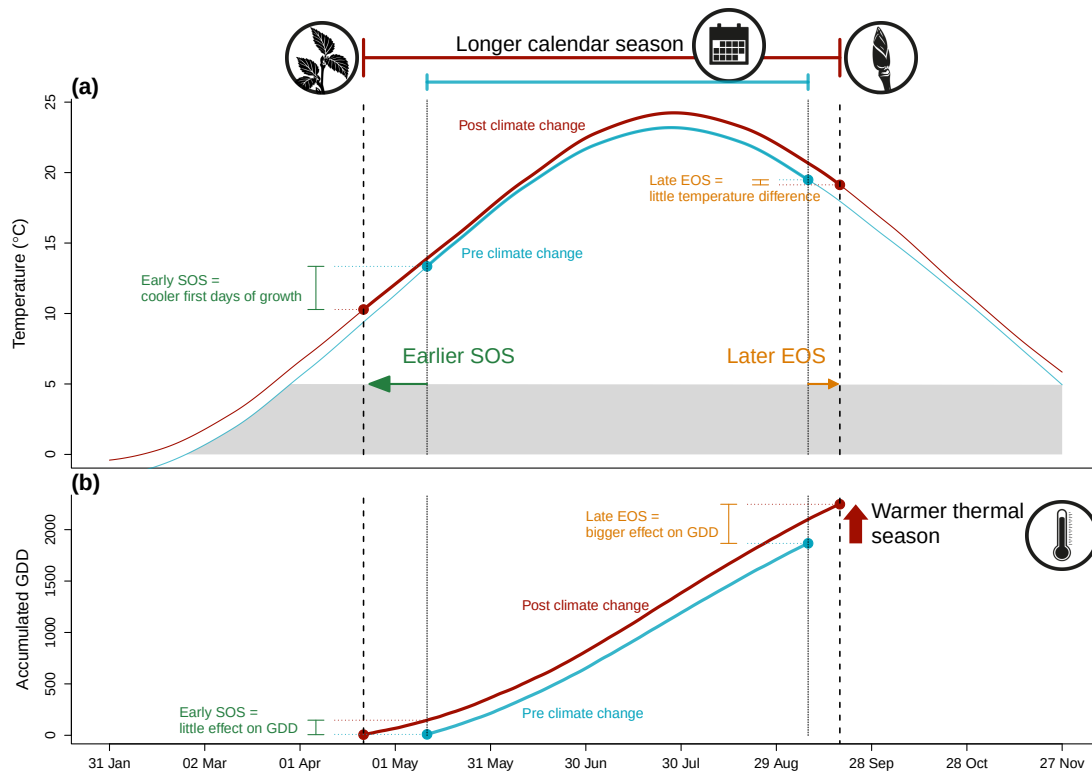


Figure 1.1: We show how shifting growing season length with climate change affects both the temperature trees experience throughout the growing season (a), and how much growing degree days (GDD) they accumulate (b). We used mean daily temperature data from the Boston airport Climate station for the smoothed temperature and GDD curves. We represent the changes in temperature before (1945-1965 in blue) and after the 1980s increase in climate warming (2005-2025 in red). With climate change, spring events (green arrow) have advanced and fall phenology (orange arrow) has delayed, but to much lesser extent (day advancements are fake and represented by the arrows), this lengthened the calendar growing season, but concurrently changed the temperature trees experience early in their season. Growth onset earlier in the season means the first days of growth are cooler (a) resulting in little accumulation of GDD (b). Though shifts in EOS are weaker and the temperature experienced changed little (a), the GDD accumulation rate late in the season is higher than early in the season. Alongside warmer summer temperature, this results in higher GDD accumulation, warming their thermal seasons.

Results

1. Does tree growth increase with longer growing seasons?
2. Did an earlier start and later end of season increase growth?
3. Does tree growth increase with longer growing seasons and is it consistent across species and provenances?
4. How did growth change across years, and how did it relate to climate?

1. Summary basic information

- (a) Wildchrokie includes four deciduous tree species from four sites with leaf phenology data spanning three years (2018 to 2020) for a total of 239 leafout and 230 budset observations, yielding 227 complete growing seasons (GS) for 81 individuals.
- (b) Mean summer temperature were similar across years with 2019 being the coolest (22.38°C), 2020 the hottest (23.07°C), and 2018 in between (22.85°C)
- (c) The calendar and thermal seasons ranged from 90 days and 1465.1 GDD (*B. alleghaniensis* in 2018) to 173 days and 2513.6 GDD (*A. incana* in 2018)
- (d) The earliest leafout day of year was 22-Apr (*A. incana* in 2019) and the latest, 05-Jun (also *A. incana*, but in 2020). The earliest budset day of year was 13-Aug (*B. alleghaniensis* in 2018) and the latest, 24-Oct (*A. incana* in 2018)
- (e) No drought for the three years, except for in 2020, where a moderate summer drought ramped up to a severe drought in September.
- (f) Annual ring widths across years varied from 0.39 mm (*B. alleghaniensis* in 2020) to 13.55 mm (*B. populifolia* in 2018) (Figure S5).

2. Does tree growth increase with longer growing seasons?

- (a) Longer growing seasons increased growth for all species, but varied depending on the metric used to measure it.
- (b) Longer thermal season increased growth for *B. papyrifera* ($\beta = 0.15$, UI_{90} [0.06, 0.23]), *B. populifolia* ($\beta = 0.07$, UI_{90} [0.02, 0.12]), and *B. alleghaniensis* ($\beta = 0.04$, UI_{90} [0, 0.08]), but *A. incana* had a weak effect, but trended in the same direction ($\beta = 0.02$, UI_{90} [-0.01, 0.05]) (Figure 1.2a and e) (TableS1).
- (c) *A. incana* grew more with longer calendar seasons ($\beta = 0.05$, UI_{90} [0, 0.1]), but of the three species that responded to thermal season, only *B. papyrifera* also grew more with longer calendar seasons ($\beta = 0.18$, UI_{90} [0.05, 0.31]).
- (d) *B. alleghaniensis* ($\beta = 0.03$, UI_{90} [-0.03, 0.09]), and *B. populifolia* ($\beta = 0.05$, UI_{90} [-0.02, 0.13]) trended in the same direction (Figure 1.2b and f) (Table S1).

- (e) Results with basal area increment (BAI) for the GDD model were broadly similar to ring width (see Table S4)
 - (f) On a scaled axis, both predictors were similarly predictive (FigureS10).
3. Did a later start and end of season increase growth?
- (a) A longer calendar season increased growth for two species, but only one grew more with an earlier start of season (*A. incana*: $\beta = -0.07$, UI_{90} [-0.13, -0.01]) (Figure 1.2c and g) (Table S1).
 - (b) *B. alleghaniensis* which did not grow more with longer calendar season, grew less with an earlier start of season, unlike our prediction ($\beta = 0.1$, UI_{90} [0, 0.2]) (Figure 1.2c and g) (Table S1). git statusgit status
 - (c) Three species grew more with a later end of season (*B. alleghaniensis*: $\beta = 0.12$, UI_{90} [0.05, 0.18], *B. populifolia*: $\beta = 0.1$, UI_{90} [0.02, 0.17]), with *B. papyrifera* ($\beta = 0.21$, UI_{90} [0.09, 0.33]) being the only one of those three that also grew more with longer calendar season (Figure 1.2d and h) (Table S1).
 - (d) The estimates were unchanged when using the full (SOS $n = 239$ and $n = 230$) or restricted datasets ($n = 227$) (Figure S1a).
4. How did tree growth change across years, provenances and tree ids?
- (a) We did not find any clear effect of year on growth (Figure ??) and there is no clear differences of growth looking across species (Figure S2 and Figure S3)
 - (b) Growth did not change across any provenances (Figure 1.3) (Table S2), and the variation across sites ($\sigma = 0.17$, UI_{90} [0.02, 0.48]) and tree ids ($\sigma = 0.17$, UI_{90} [0.05, 0.27]) was similar

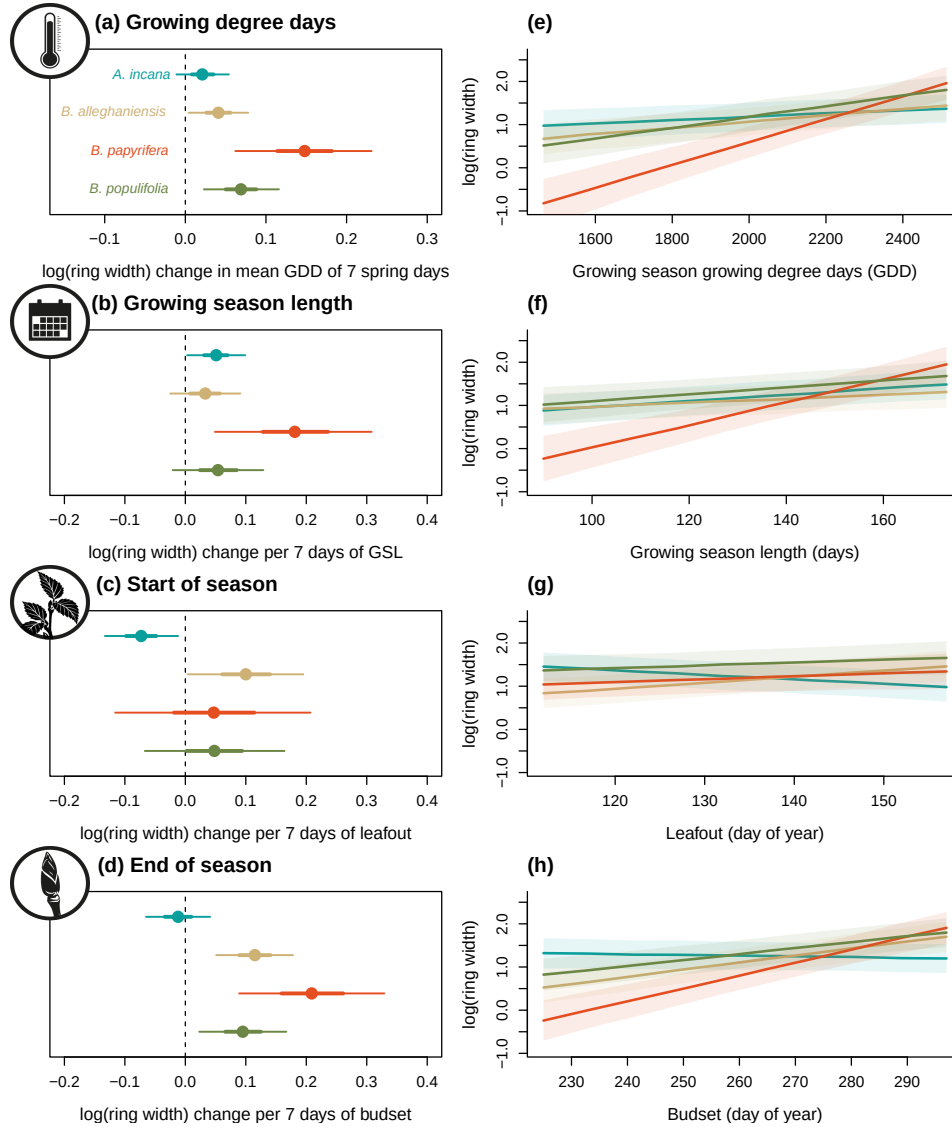


Figure 1.2: Model slope estimates representing ring width change as a function of four different predictors across four species. GDD represent the accumulated growing degree days between leafout and budset and the models estimates shows ring width change per 7 average spring days GDD (55.73) (a). GSL is the number of days between leafout and budset (b). SOS represent the start of season, which is leafout (c) and EOS, the end of season through budset (d). The dots represent the mean of each species and the lines, the 50% and 90% uncertainty intervals (UI).

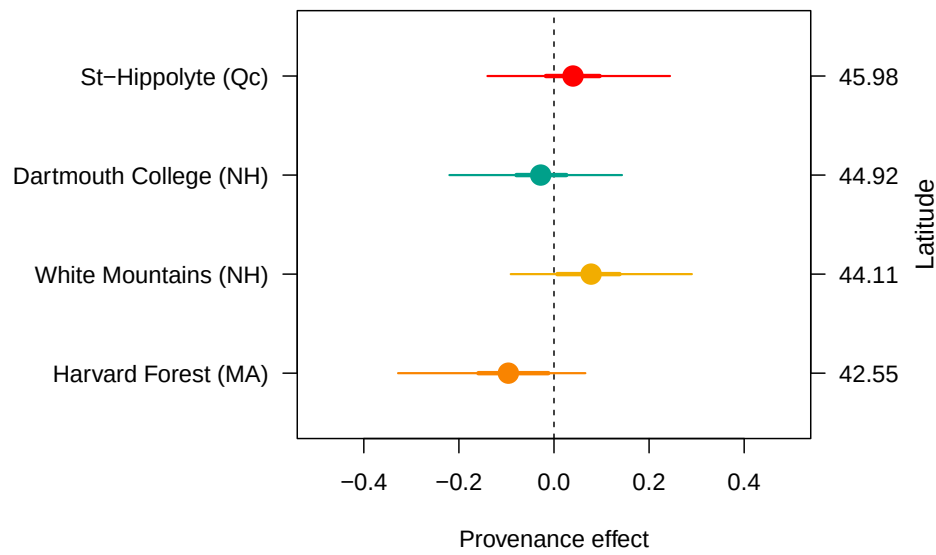


Figure 1.3: The effect of provenances on growth across a latitudinal gradient. (a) Model intercept parameters for each site with growing degree days (GDD) as the predictor (effects were similar for the other predictors, see Table S2). The dots represent the mean of each provenance and the thick lines, the 50% and 90% uncertainty intervals (UI). The sites are color-coded from (b), the map showing their locations.

Discussion

1. (a) Used 81 juvenile trees of 4 species from 4 provenances to understand if longer GS increased growth
 - i. We used annual growth with tree rings and leaf phenology for SOS and EOS
 - ii. Longer GS increased growth for all species, but varied depending on the metric
 - iii. Thermal season was a better predictor for growth than the calendar season
 - iv. Little effect of provenance
2. The thermal season better predicts growth than the calendar season
 - (a) Thermal is a better predictor of growth because it's more closely aligned to when and how fast trees grow
 - i. Thermal weights growth days better than calendar season
 - ii. Cool spring days count for less for thermal season vs calendar season (Figure 1.1)
 - iii. Stronger response to EOS than SOS, because temperature is warmer = more GDD (Figure 1.1) (Buonaiuto, unpublished)
 - (b) SOS and EOS to disentangle the calendar season effect on growth
 - i. Early SOS didn't increase growth for most species (like Dow 2022)
 - ii. Studies that solely focus on SOS will miss that longer seasons may lead to more growth
 - iii. Late EOS increased growth for most species; relates to GDD, less to GSL
 - (c) Future response of growth in response to soil moisture is uncertain
 - i. Later EOS coupled with appropriate soil moisture = growth increase—as we and others found
 - ii. No current increased drought in our species native range = growth would increase
 - iii. Positive effect of later EOS offset if soil moisture becomes inadequate
 - iv. While 2020 had a late of season drought, late of season droughts in the north east are rare—could change with warmer conditions
 - v. Other temperate regions have growth decrease attributed to reduced soil moisture
3. Contrasting growth responses across species, but not across provenances
 - (a) Species responded differently to different season predictors
 - i. *A. incana* grow more with GSL and earlier SOS: most growth happens in spring
 - ii. *A. incana* is ruderal, takes advantage of cool and short days in spring
 - iii. Contrary, *B. papyrifera* and *B. populifolia* driven by EOS: more reactive to warmer, late season temperature
 - iv. *B. alleghaniensis* SOS and EOS trends counterinteract = no response to GSL
 - (b) Growth strategies across species may change how responsive they are

- i. All our species are either fast or moderately fast growing species—known to track changing conditions better
 - ii. This could change if we used slow growing species
 - iii. Juvenile stages are known to be more reactive, could explain why they grew more with warmer seasons
 - iv. These trees in nature would be below canopy, so they would have much less light and might not have responded positively
- (c) No effect of provenance suggests local adaption effect on growth is weak under current conditions
 - i. Good understanding of which leaf phenophase is locally adapted: budset > leafout
 - ii. Little evidence of local adaptation on growth may be caused by little local adaptation on budset
 - iii. We know little about the effect of local adaptation on growth
 - iv. Understanding this requires teasing out the effects of local adaptation on leaf and wood phenology at the SOS and EOS
- 4. Future carbon models and range shift projections need to consider species
 - (a) What we did is important
 - i. Juvenile trees may be more responsive, but they would be competing for light in nature
 - ii. EOS and SOS asymmetrically matter to growth and we don't understand why
 - iii. Highlights importance of considering not just SOS (e.g. Dow, Wheeler2016Snow)
 - iv. Closely related species trends differed; they matter and we don't consider them enough

Supplemental results

Table S1: Species level slope parameters (mean and 90% uncertainty intervals, in square brackets) across the four different predictors for Wildchrokia. GDD: growing degree days, GSL: growing season length, SOS: start of season and EOS: end of season.

Species	GDD	GSL	SOS	EOS
<i>A. incana</i>	0.02 [-0.01, 0.05]	0.05 [0.00, 0.10]	-0.07 [-0.13, -0.01]	-0.01 [-0.07, 0.04]
<i>B. alleghaniensis</i>	0.04 [0.00, 0.08]	0.03 [-0.03, 0.09]	0.10 [0.00, 0.20]	0.12 [0.05, 0.18]
<i>B. papyrifera</i>	0.15 [0.06, 0.23]	0.18 [0.05, 0.31]	0.05 [-0.12, 0.21]	0.21 [0.09, 0.33]
<i>B. populifolia</i>	0.07 [0.02, 0.12]	0.05 [-0.02, 0.13]	0.05 [-0.07, 0.16]	0.10 [0.02, 0.17]

Table S2: Site level intercept parameters (mean and 90% uncertainty intervals, in square brackets) across the four different predictors for Wildchrokia. GDD: growing degree days, GSL: growing season length, SOS: start of season and EOS: end of season.

Site	GDD	GSL	SOS	EOS	Lon.
Harvard Forest (MA)	-0.10 [-0.33, 0.07]	-0.09 [-0.34, 0.09]	-0.04 [-0.26, 0.12]	-0.10 [-0.33, 0.07]	-72.19
White Mountains (NH)	0.08 [-0.09, 0.29]	0.08 [-0.09, 0.31]	0.08 [-0.08, 0.29]	0.07 [-0.09, 0.27]	-71.23
Dartmouth College (NH)	-0.03 [-0.22, 0.14]	-0.04 [-0.25, 0.15]	-0.06 [-0.27, 0.10]	-0.04 [-0.24, 0.12]	-71.15
St-Hippolyte (Qc)	0.04 [-0.14, 0.24]	0.04 [-0.16, 0.26]	0.02 [-0.15, 0.22]	0.04 [-0.14, 0.24]	-74.03

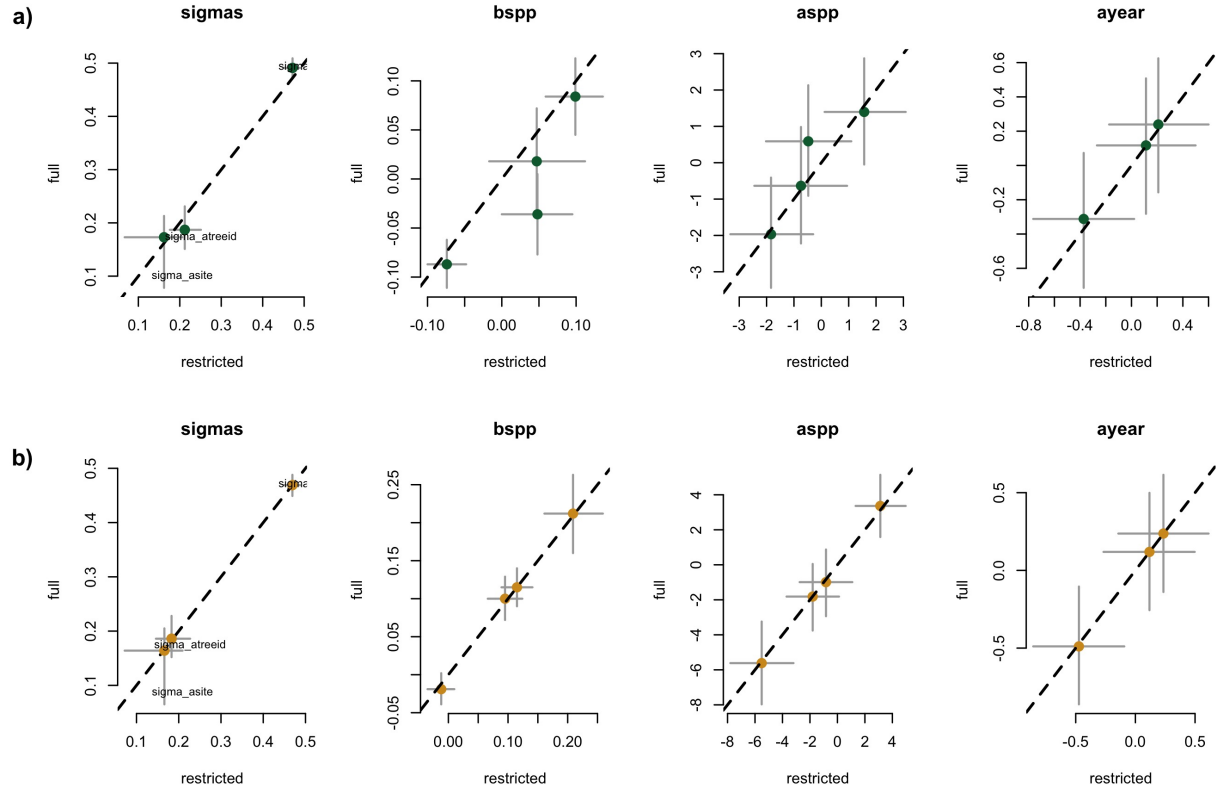


Figure S1: Full vs most restricted rows of data for the model with SOS (a) EOS (b) as the predictor for Wildchrokie. Each panel represent the different parameters used to fit the models.

Table S3: Posterior means and 90% uncertainty intervals for all model parameters across predictors

Parameter	GDD	GSL	SOS	EOS
$\sigma_{\alpha_{tree}}$	0.17 [0.05, 0.27]	0.18 [0.06, 0.28]	0.21 [0.10, 0.30]	0.18 [0.06, 0.28]
$\sigma_{\alpha_{site}}$	0.17 [0.02, 0.47]	0.18 [0.02, 0.48]	0.17 [0.02, 0.46]	0.17 [0.02, 0.46]
σ_y	0.48 [0.43, 0.53]	0.48 [0.44, 0.53]	0.47 [0.43, 0.52]	0.47 [0.42, 0.52]
$\beta_{sppA.incana}$	0.02 [-0.01, 0.05]	0.05 [0.00, 0.10]	-0.07 [-0.13, -0.01]	-0.01 [-0.07, 0.04]
$\beta_{sppB.allegghaniensis}$	0.04 [0.00, 0.08]	0.03 [-0.03, 0.09]	0.10 [0.00, 0.20]	0.12 [0.05, 0.18]
$\beta_{sppB.papyrifera}$	0.15 [0.06, 0.23]	0.18 [0.05, 0.31]	0.05 [-0.12, 0.21]	0.21 [0.09, 0.33]
$\beta_{sppB.populifolia}$	0.07 [0.02, 0.12]	0.05 [-0.02, 0.13]	0.05 [-0.07, 0.16]	0.10 [0.02, 0.17]
$\alpha_{siteDartmouthCollege(NH)}$	-0.03 [-0.22, 0.14]	-0.04 [-0.25, 0.15]	-0.06 [-0.27, 0.10]	-0.04 [-0.24, 0.12]
$\alpha_{siteHarvardForest(MA)}$	-0.10 [-0.33, 0.07]	-0.09 [-0.34, 0.09]	-0.04 [-0.26, 0.12]	-0.10 [-0.33, 0.07]
$\alpha_{siteSt-Hippolyte(Qc)}$	0.04 [-0.14, 0.24]	0.04 [-0.16, 0.26]	0.02 [-0.15, 0.22]	0.04 [-0.14, 0.24]
$\alpha_{siteWhiteMountains(NH)}$	0.08 [-0.09, 0.29]	0.08 [-0.09, 0.31]	0.08 [-0.08, 0.29]	0.07 [-0.09, 0.27]
$\alpha_{year2018}$	0.19 [-0.77, 1.15]	0.24 [-0.70, 1.17]	0.20 [-0.75, 1.14]	0.22 [-0.71, 1.16]
$\alpha_{year2019}$	0.18 [-0.79, 1.14]	0.10 [-0.83, 1.06]	0.11 [-0.84, 1.04]	0.11 [-0.83, 1.05]
$\alpha_{year2020}$	-0.43 [-1.40, 0.53]	-0.34 [-1.27, 0.60]	-0.38 [-1.32, 0.56]	-0.49 [-1.42, 0.46]
$\alpha_{sppA.incana}$	1.06 [-3.19, 5.21]	0.11 [-3.92, 4.21]	1.48 [-2.13, 5.13]	3.06 [-1.40, 7.36]
$\alpha_{sppB.allegghaniensis}$	0.21 [-3.97, 4.34]	0.35 [-3.69, 4.53]	-1.91 [-5.64, 1.79]	-1.85 [-6.39, 2.59]
$\alpha_{sppB.papyrifera}$	-4.00 [-8.89, 0.75]	-2.61 [-7.17, 1.95]	-0.79 [-5.03, 3.36]	-5.57 [-11.02, -0.07]
$\alpha_{sppB.populifolia}$	-0.66 [-4.94, 3.63]	0.22 [-3.97, 4.45]	-0.53 [-4.29, 3.22]	-0.88 [-5.38, 3.67]

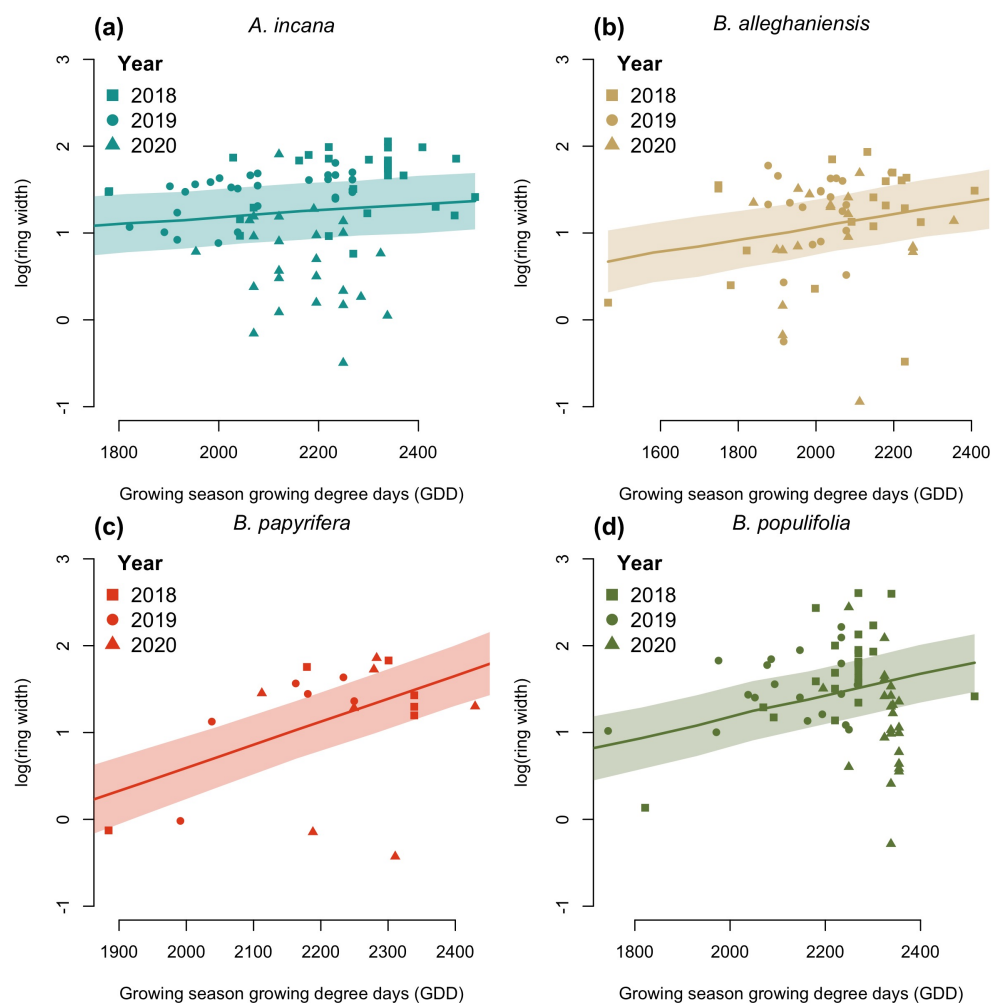


Figure S2: Ring width trends in response to thermal season (GDD). The plots are the reconstructed species estimates posterior distribution with error. The lines represent the mean of the posterior distribution and the shaded area, the 50% UI. The dots are the empirical data shaped by year.

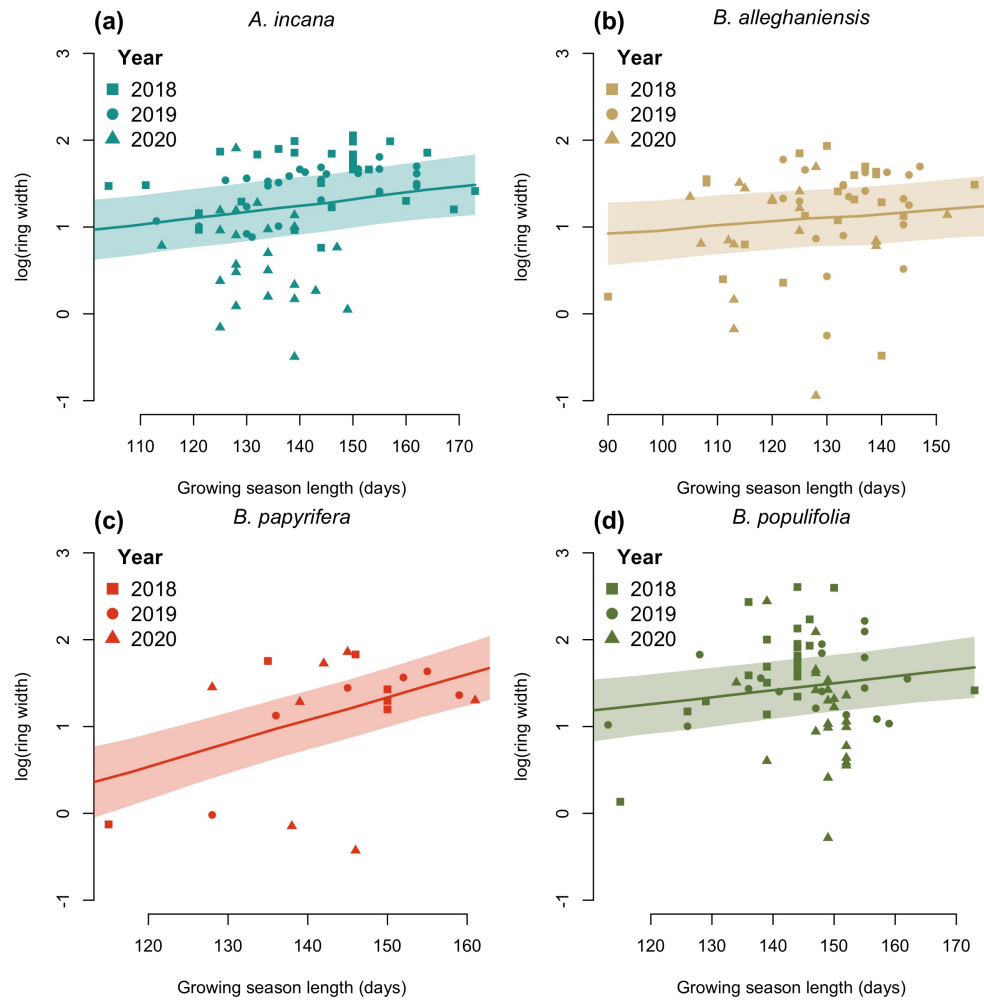


Figure S3: Ring width trends in response to calendar season (GSL). The plots are the reconstructed species estimates posterior distribution with error. The lines represent the mean of the posterior distribution and the shaded area, the 50% UI. The dots are the empirical data shaped by year.

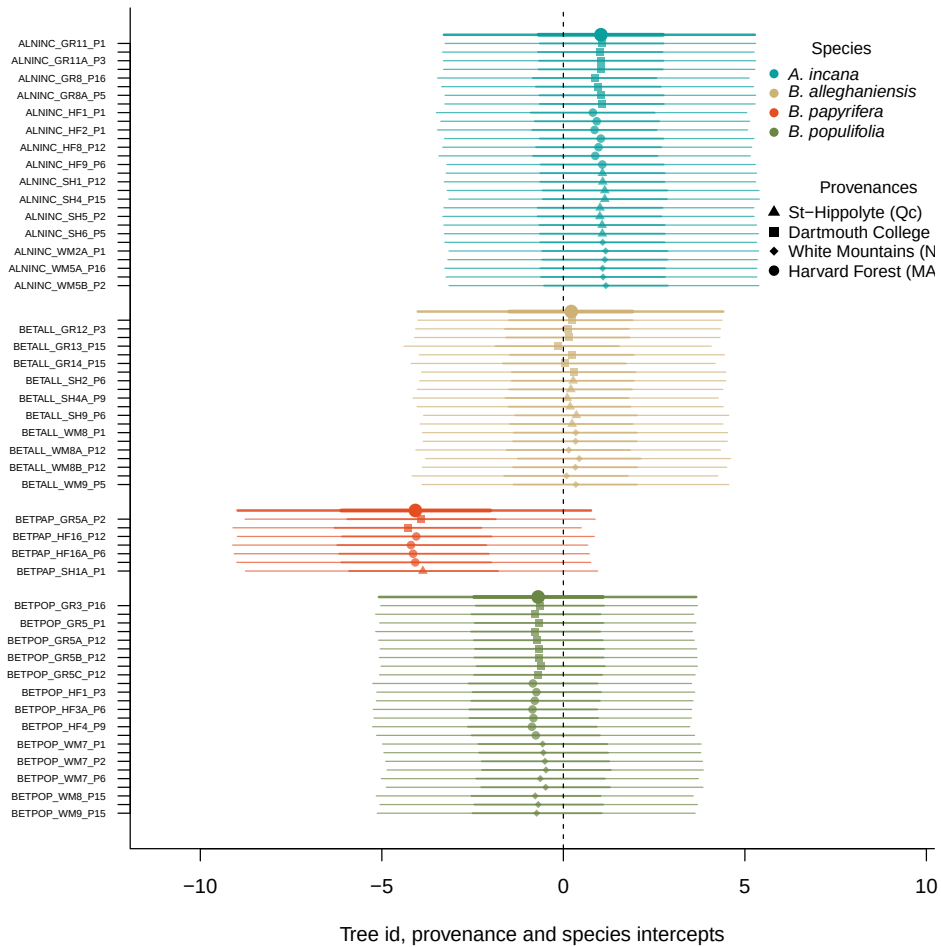


Figure S4: Intercept estimates for each group of the GDD model. The different tree id values represent the combined intercepts of tree id, provenance, species and averaged year intercepts. The bigger dots and lines represent the species intercepts which are the species level estimates averaged across all of their tree ids. The larger dots and lines represent aggregated average across these values. The dots are the mean of the posterior distribution and the thick and thin lines, the 50 and 90% UI.

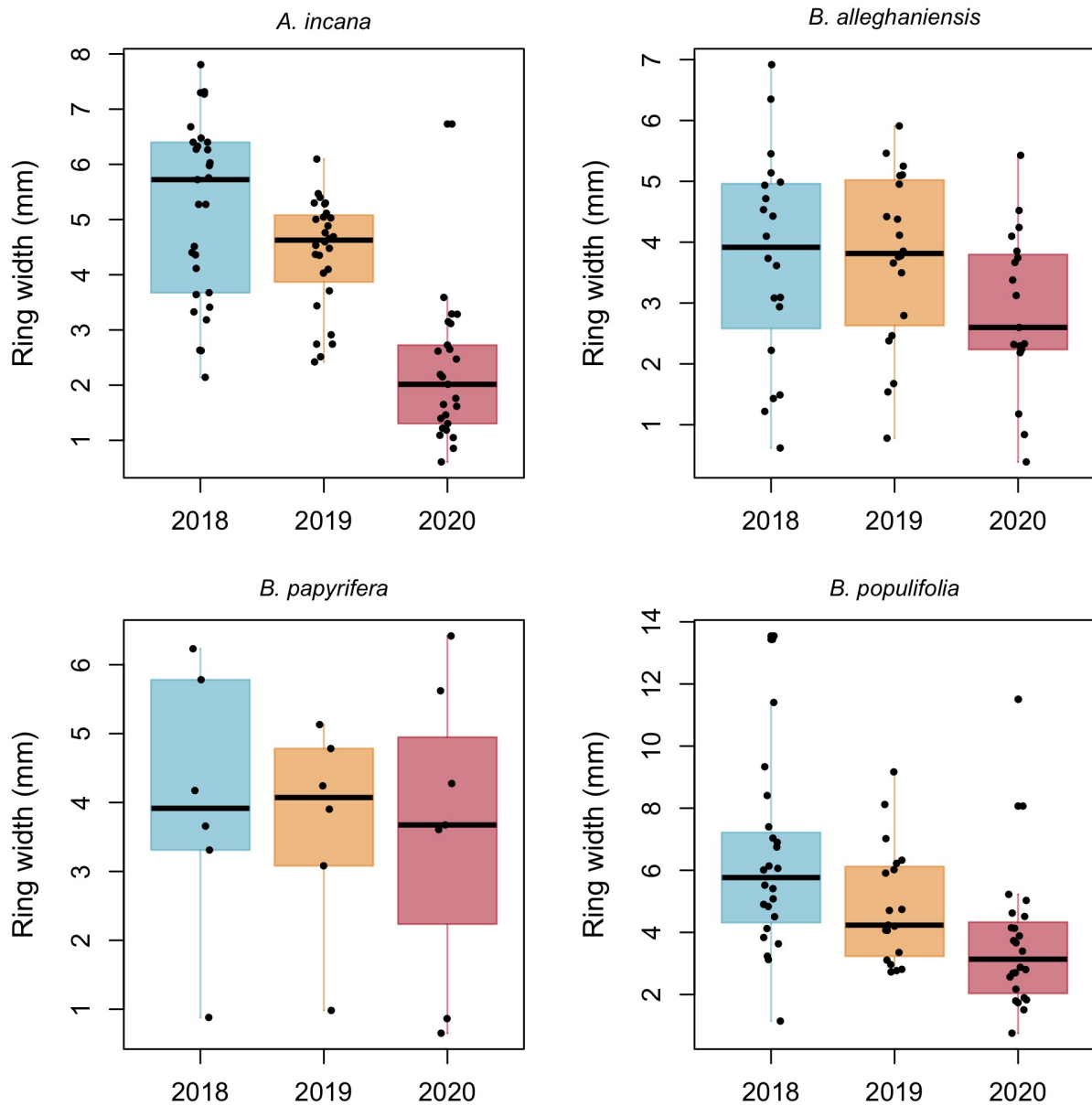


Figure S5: Boxplot representing the empirical data of our ring widths (mm) faceted by species. The dots represent the raw data, the line the mean and the box the IQR.

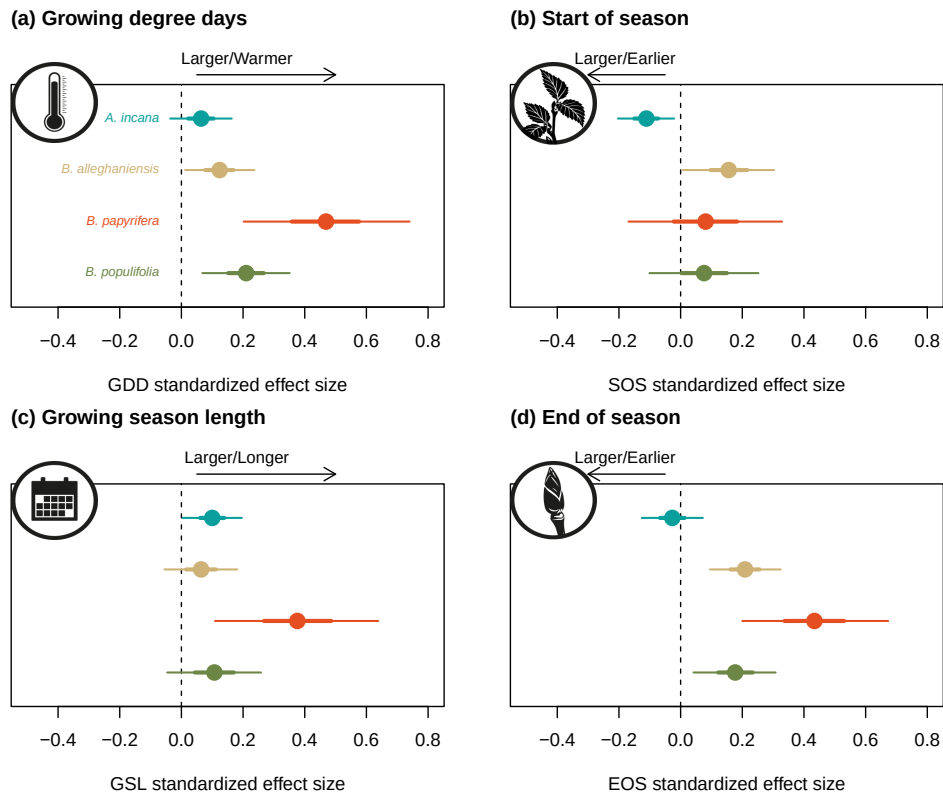


Figure S6: Standardized (z-scored) slope estimates for Wildchrokie. The dots represent the mean of the posterior distribution, the thick lines the 50% uncertainty intervals (UI) and the thinner lines, the 90% UI.

Table S4: Posterior means and 90% uncertainty intervals for all model parameters, when using basal area increment (BAI) as the response variable

Parameter	Estimate
$\sigma_{\alpha_{tree}}$	0.52 [0.41, 0.64]
$\sigma_{\alpha_{site}}$	0.27 [0.03, 0.73]
σ_y	0.63 [0.57, 0.70]
$\beta_{sppA.incana}$	0.11 [-0.03, 0.25]
$\beta_{sppB.alleganiensis}$	0.20 [0.02, 0.36]
$\beta_{sppB.papyrifera}$	0.92 [0.49, 1.34]
$\beta_{sppB.populifolia}$	0.41 [0.20, 0.61]
$\alpha_{siteDartmouthCollege(NH)}$	-0.11 [-0.47, 0.17]
$\alpha_{siteHarvardForest(MA)}$	-0.09 [-0.45, 0.19]
$\alpha_{siteSt-Hippolyte(Qc)}$	0.06 [-0.23, 0.42]
$\alpha_{siteWhiteMountains(NH)}$	0.15 [-0.11, 0.52]
$\alpha_{year2018}$	-0.36 [-1.32, 0.61]
$\alpha_{year2019}$	0.33 [-0.65, 1.32]
$\alpha_{year2020}$	-0.04 [-1.01, 0.94]
$\alpha_{sppA.incana}$	3.23 [-2.25, 8.55]
$\alpha_{sppB.alleganiensis}$	2.09 [-3.40, 7.54]
$\alpha_{sppB.papyrifera}$	-7.59 [-14.84, -0.29]
$\alpha_{sppB.populifolia}$	-0.07 [-5.98, 5.57]

Chapter 2

Coring Treepotters

Introduction

1. Climate change shift plant phenology
 - (a) Earlier spring and delayed autumn lengthens the growing season
 - (b) With longer seasons, trees should grow more
 - (c) Temperate forests offset a large portion of human CO₂ emissions
 - (d) Most of that carbon is stored in mature, canopy trees so they are important
2. Increased growth with longer seasons may not happen
 - (a) Some new evidence shows trees don't grow more with longer seasons
 - (b) This could mislead carbon models
 - (c) Highlights the need to understand why trees might not grow more
3. Growth strategies (i.e. acquisitive vs conservative) and wood anatomy may guide species responses to GS
 - (a) Diffuse vs ring porous species start growth at different time—experience different temperature throughout the season
 - (b) Longer seasons may benefit slow growth species > fast growth species
 - (c) Different carbon allocation strategies may mute short term growth responses
4. Previous year conditions can affect the following year's growth
 - (a) Deciduous trees use stored carbon at the beginning of the season
 - (b) Mature trees especially are known to rely on stored carbon for early growth
 - (c) Thus, effect of environmental conditions on yearly growth can be blurred by a multi-year lag effect

5. Look at growth and GSL relationship in an urban arboretum where drought and competition are limited
 - (a) GS number of days vs temperature conditions
 - (b) Growth with tree rings and SOS and EOS with leaf phenology from citizen scientists
 - (c) Will look at 11 species of mature trees across nine years.

Methods

We did a bunch of stuff that I am not outlining. Phenophases by (Denny *et al.*, 2014)

Results

1. Summary basic information

- (a) Dataset includes 11 deciduous tree species with leaf phenology data spanning nine years (2016 to 2024) for a total of 321 leafout and 284 leaf coloring observations, yielding 278 complete growing seasons (GS) for 49 individuals.
- (b) Mean summer temperatures spanned 21.53°C (2017) to 23.07°C (2020), with an average of 22.51°C across the nine years.
- (c) 2020 and 2022 had moderate droughts in the summer (June, July and August) ramped up to severe in September. A moderate drought in June 2016, ramped up to severe in July, then extreme in August and September.
- (d) GSL ranged from 77 days (*A. flava* in 2018) to 194 days (*A. saccharum* in 2021)
- (e) GS thermal sum ranged from 1048.8 GDD (*A. flava* in 2020) to 2634.7 GDD (*T. americana* in 2016)
- (f) The earliest leafout day of year was 08-Apr (*A. flava* in 2021) and the latest, 15-May (*C. ovata* in 2020). The earliest leaf coloring day of year was 15-Jul (*A. flava* in 2021—same individual with earliest leafout) and the latest, 06-Nov (*T. americana* in 2016)
- (g) Annual ring widths across years varied from 0.43 mm (*A. saccharum* in 2019) to 11.75 mm (*T. americana* in 2019) (Figure S2).

2. Does tree growth increase with longer growing seasons?

- (a) No clear trend of growing season length effect across our eleven species and season predictors.
- (b) Results reported as means and 90% uncertainty intervals (UI) that correspond to ring width change in mm on the log scale—henceforth "RW" per scaled predictor (one week (seven days) of average spring growing degree days (GDD); (58.32) for the thermal season, one week of growing season length (GSL) for the calendar season, one week of leafout for the start of season (SOS) and one week of budset for the end of season (EOS).
- (c) Longer thermal season did not increase growth for any species and decreased growth for *B. alleghaniensis* (-0.08, UI: -0.12, -0.03 RW change per scaled GDD; Figure S1a and e; Table S1).
- (d) No species responded to longer calendar season (Figure S1b and f; Table S1).
- (e) On a scaled axis (z-scored), both GDD and GSL were similarly predictive (Figure S10)

3. Does a warmer thermal season increase growth in the following year?

- (a) Previous year GDD increased growth for *B. alleghaniensis* (0.15, UI: 0.02, 0.28 RW change per scaled GDD), and decreased growth for *B. nigra* (-0.08, UI: -0.17, 0 RW change per scaled GDD; Figure S2b)

- (b) Interestingly, *B. nigra* did not solely respond to current year's conditions (Figure S1a and e; Table S1), but grew more with warmer thermal season when the previous conditions were considered (0.12, UI: 0.03, 0.21 RW change per scaled GDD; Figure S2a).
4. Did an earlier start and later end of season increase growth?
- (a) An earlier start of season did not increase growth for any species and decreased growth for *T. americana* (0.15, UI: 0.02, 0.27 RW change per seven SOS days; Figure S1c and g; Table S1), which did not respond to the thermal or the calendar season.
 - (b) A later end of season did not increase growth for any species and decreased growth for *B. al-leghaniensis* (-0.09, UI: -0.16, -0.01 RW change per seven EOS days; Figure S1d and h; Table S1), which also decreased growth with warmer thermal season.
 - (c) The estimates were unchanged when using the full or restricted datasets (SOS $n = 321$ and EOS $n = 284$ vs $n = 278$; Figure S8b), or when using budburst instead of leafout as the start of the growing season (Figure S9c).
 - (d) A later SOS delayed EOS only for *P. deltoides* (1.04, UI: 0.39, 1.7 EOS change in days per SOS day; Figure S11)
5. How did tree growth change across years and tree ids?
- (a) We did not find any clear effect of year on growth (Figure S1) and no clear differences of growth across species (Figure S5 and Figure S6)
 - (b) Growth was similar across individual trees (Figure S7).

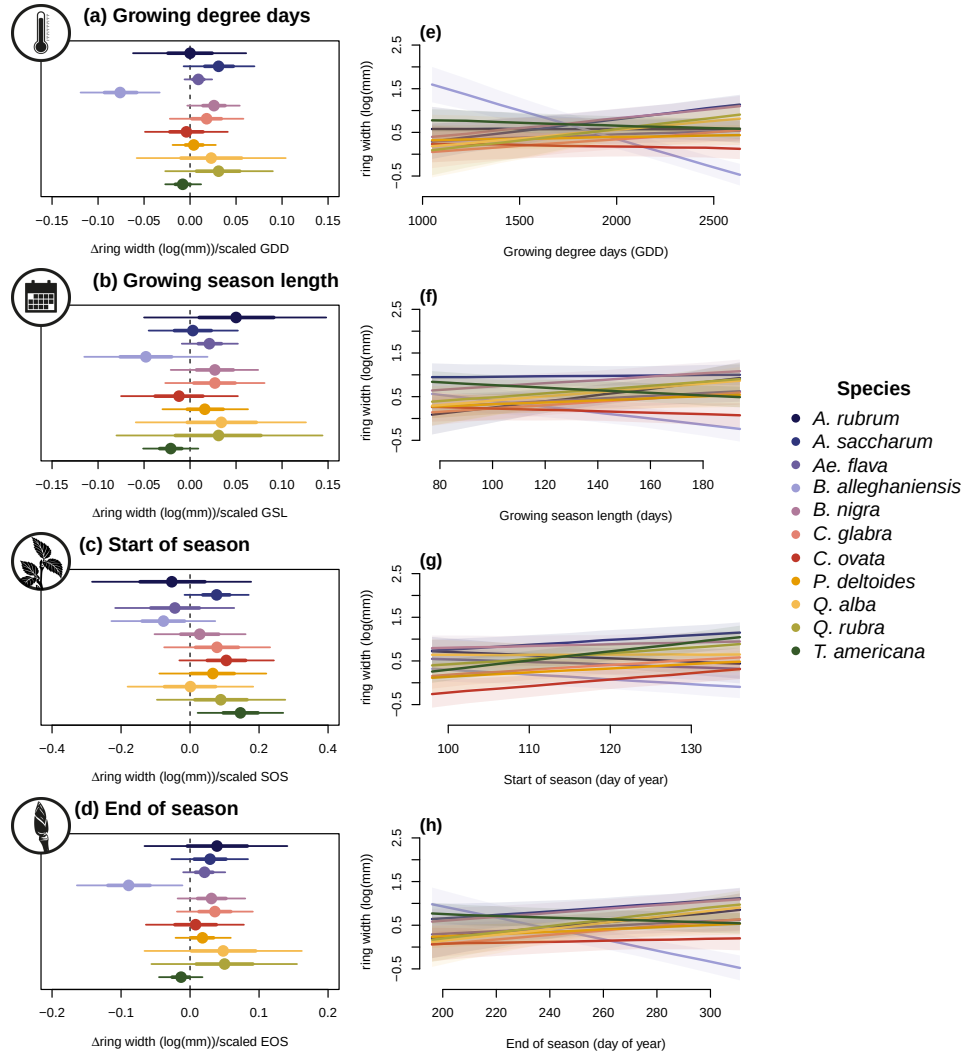
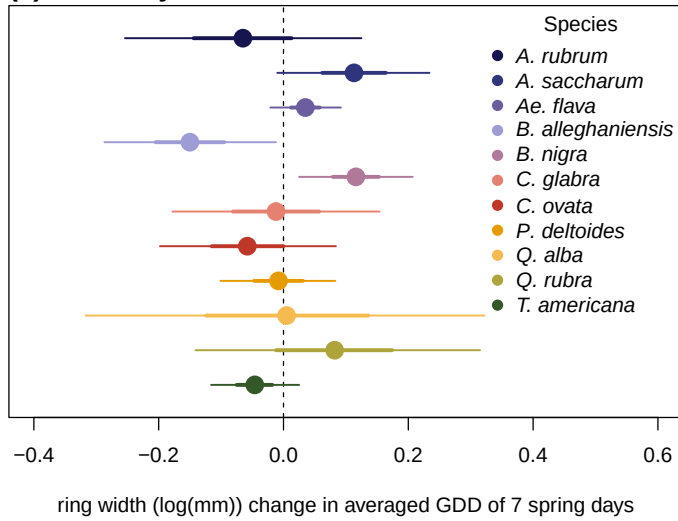


Figure S1: Estimates of ring width change as a function of four different predictors (growing degree days, GDD (mean daily temperature above 5°C); growing season length, GSL; start of season, SOS; end of season, EOS) across our eleven species. The species colors are the same throughout the figures. Panels (a) to (d) represent the model slope estimates for each species, for each predictor, which we represent through ring width change on the log scale per seven average spring days GDD (58.32 GDD; panel a), seven days of GSL (panel b), seven days of leafout (SOS; panel c) and seven days of leaf colouring (EOS; panel d). The dots represent the mean of each species and the lines, the 50% and 90% uncertainty intervals (UI; see Table S1 for parameter estimates). Panels (e) to (h) represent the simulated response curves of ring width as a function of each predictor, with the lines depicting the mean response at each simulated predictor value, and the shaded area, the interquartile ranges (25% and 75% UI). These model do not include a slope on the previous year GDD (see Figure S2 for these estimates).

(a) Current year



(b) Previous year

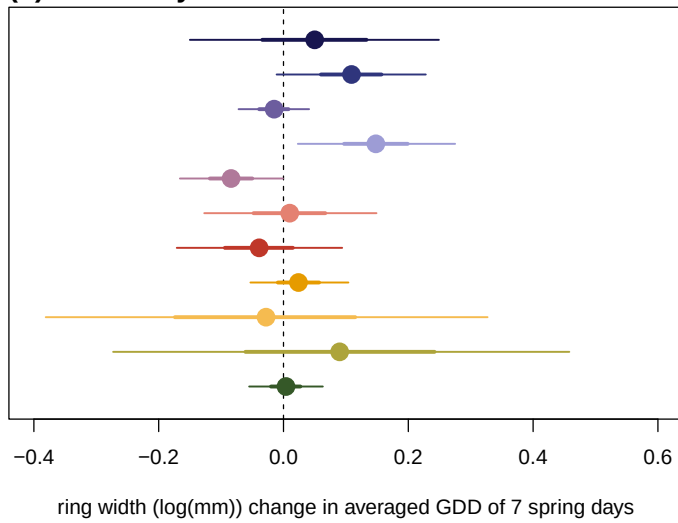


Figure S2: Model slope estimates representing ring width change as a function of the current year growing degree days (GDD; a) and the previous year GDD (b). GDD represent the accumulated growing degree days between leafout and leaf colouring and the models estimates shows ring width change (on the log scale) per seven average spring days GDD (58.32; see data analyses for more details). The dots represent the mean of each species and the lines, the 50% and 90% uncertainty intervals (UI).

Discussion

1. Used 49 trees of 11 species to understand if mature trees grew more with longer GS
 - (a) Annual growth with tree rings and leaf phenology for SOS and EOS
 - (b) Warmer and longer GS did not increase growth for any species and decreased growth for one
 - (c) Previous year conditions shifted growth trend for two species
 - (d) Most species did not respond to SOS or EOS
2. Despite varying growth strategies, no consistent growth trend
 - (a) Thermal or calendar season had no effect on any species except for one
 - (b) *B. alleghaniensis* decrease in growth reveal that this species growth may be inhibited by warmer seasons
 - (c) *B. alleghaniensis* response to thermal season is consequential with more growth with earlier EOS: period of high GDD accumulation
 - (d) *T. americana*: grows less with earlier SOS; shows cool early spring days may reduce growth
 - (e) These suggest no difference of current GS condition/length between acquisitive vs conservative species
3. Previous year conditions affected growth for two *Betula*
 - (a) Diametrically opposed responses between *B. alleghaniensis* and *B. nigra*
 - (b) *B. alleghaniensis* trend suggests that warmer conditions in previous year increases growth, but current decreases
 - (c) Points towards allocation strategy that prioritize storage before growth—expected with conservative species
 - (d) *B. nigra* increases in the current year, but only when previous year is included
 - (e) This highlights the importance of considering not only the current year conditions
4. Weak responses across species corroborates previous findings on mature deciduous trees
 - (a) Our mature trees don't grow more with warmer or longer seasons
 - (b) Considering previous year condition is important, which may blur trees responses to current GS length/conditions
 - (c) GS are expected to keep increasing, but tree growth may not follow
 - (d) Temperate forest carbon sink may not keep increasing as expected

Supplemental results

Table S1: Species level slope parameters which represent ring width change (on the log scale), per scaled predictor (see data analyses section for more details). We report estimates as mean and 90% uncertainty intervals (in square brackets) across our four different predictors (GDD: growing degree days, GSL: growing season length, SOS: start of season and EOS: end of season).

Species	GDD	GSL	SOS	EOS
<i>A. rubrum</i>	0.00 [-0.06, 0.06]	0.05 [-0.05, 0.15]	-0.05 [-0.28, 0.18]	0.04 [-0.07, 0.14]
<i>A. saccharum</i>	0.03 [-0.01, 0.07]	0.00 [-0.04, 0.05]	0.08 [-0.02, 0.17]	0.03 [-0.03, 0.08]
<i>Ae. flava</i>	0.01 [-0.01, 0.02]	0.02 [-0.01, 0.05]	-0.04 [-0.22, 0.13]	0.02 [-0.01, 0.05]
<i>B. alleghaniensis</i>	-0.08 [-0.12, -0.03]	-0.05 [-0.12, 0.02]	-0.08 [-0.23, 0.07]	-0.09 [-0.16, -0.01]
<i>B. nigra</i>	0.03 [-0.00, 0.05]	0.03 [-0.02, 0.07]	0.03 [-0.10, 0.16]	0.03 [-0.02, 0.08]
<i>C. glabra</i>	0.02 [-0.02, 0.06]	0.03 [-0.03, 0.08]	0.08 [-0.07, 0.23]	0.04 [-0.02, 0.09]
<i>C. ovata</i>	-0.00 [-0.05, 0.04]	-0.01 [-0.07, 0.05]	0.10 [-0.03, 0.24]	0.01 [-0.06, 0.08]
<i>P. deltoides</i>	0.00 [-0.02, 0.03]	0.02 [-0.03, 0.06]	0.07 [-0.09, 0.22]	0.02 [-0.02, 0.06]
<i>Q. alba</i>	0.02 [-0.06, 0.10]	0.03 [-0.06, 0.13]	0.00 [-0.18, 0.18]	0.05 [-0.07, 0.16]
<i>Q. rubra</i>	0.03 [-0.03, 0.09]	0.03 [-0.08, 0.14]	0.09 [-0.10, 0.28]	0.05 [-0.06, 0.15]
<i>T. americana</i>	-0.01 [-0.03, 0.01]	-0.02 [-0.05, 0.01]	0.15 [0.02, 0.27]	-0.01 [-0.04, 0.02]

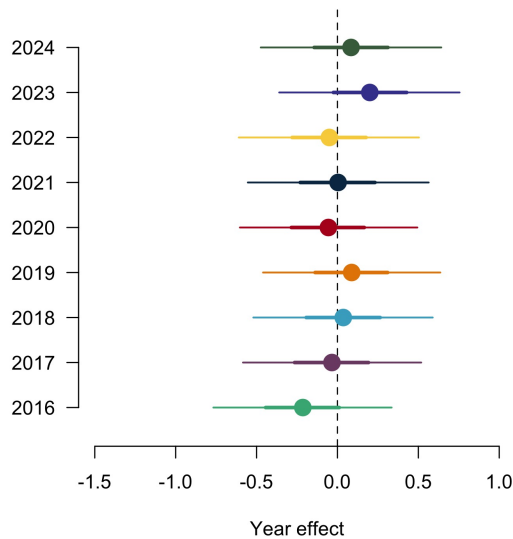


Figure S1: Ring width (on the log scale) deviations from the grand mean for each year using the GDD model (Table S2). The dots are the mean of the posterior distribution and the thick and thin lines, the 50 and 90% UI.

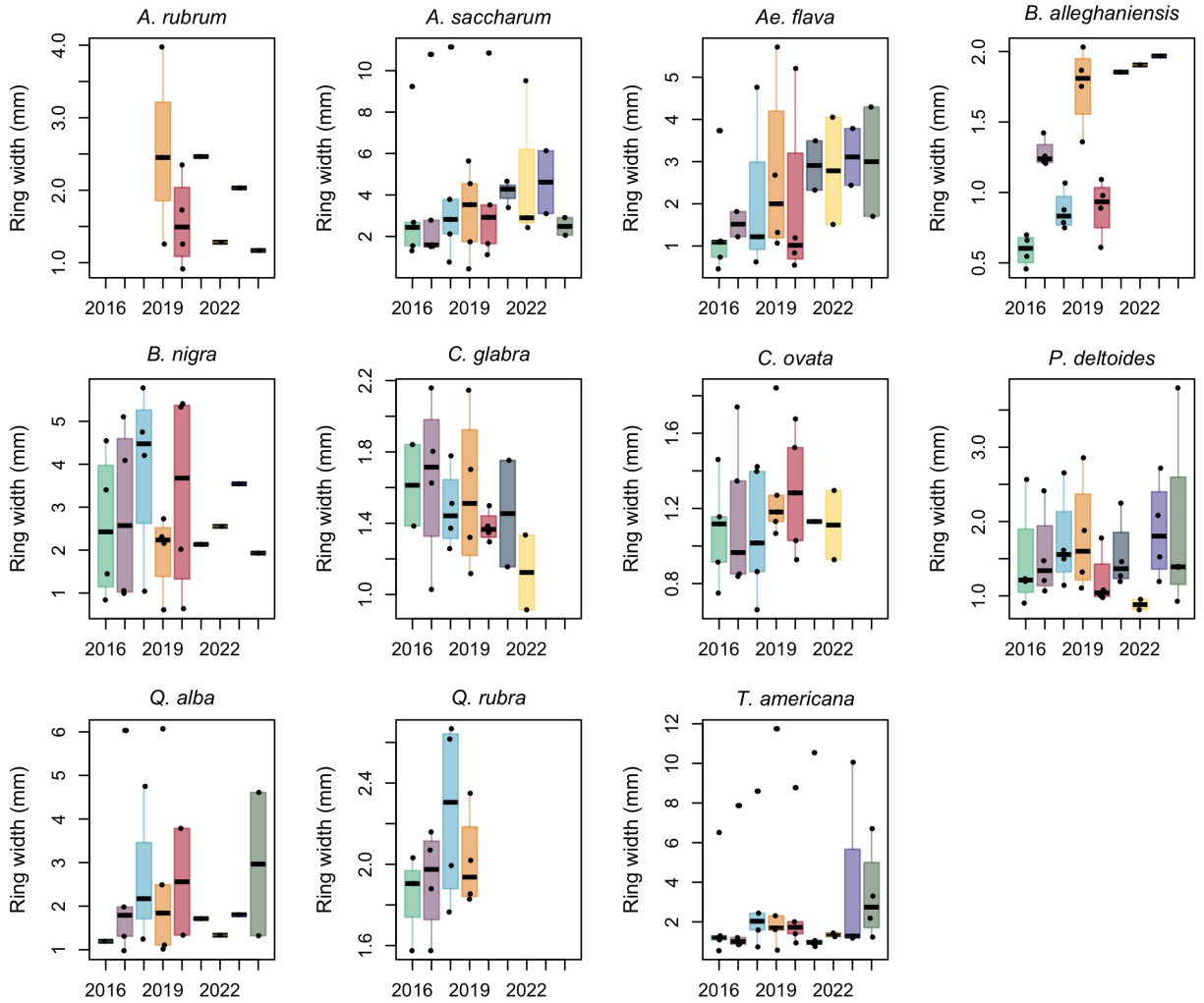


Figure S2: Boxplot representing the empirical data of our ring widths (mm) across years, faceted by species. The dots represent the raw data, the line the mean and the box the interquartile range (IQR).

Table S2: Summary output of each parameter from our four predictors, using ring width (on the log scale) as the reponse variable. We report estimates as mean and 90% uncertainty intervals (in square brackets) across our four different predictors (GDD: growing degree days, GSL: growing season length, SOS: start of season and EOS: end of season). See data analyses section for more details.

Parameter	GDD	GSL	SOS	EOS
α	0.61 [-2.08, 3.25]	0.49 [-2.09, 3.03]	0.24 [-2.39, 2.94]	0.23 [-2.48, 2.98]
$\sigma_{\alpha_{tree}}$	0.55 [0.45, 0.67]	0.55 [0.45, 0.68]	0.54 [0.44, 0.66]	0.55 [0.45, 0.67]
σ_y	0.31 [0.29, 0.34]	0.32 [0.29, 0.34]	0.32 [0.29, 0.34]	0.32 [0.29, 0.34]
$\beta_{sppA.rubrum}$	0.00 [-0.06, 0.06]	0.05 [-0.05, 0.15]	-0.05 [-0.28, 0.18]	0.04 [-0.07, 0.14]
$\beta_{sppA.saccharum}$	0.03 [-0.01, 0.07]	0.00 [-0.04, 0.05]	0.08 [-0.02, 0.17]	0.03 [-0.03, 0.08]
$\beta_{sppAe.flava}$	0.01 [-0.01, 0.02]	0.02 [-0.01, 0.05]	-0.04 [-0.22, 0.13]	0.02 [-0.01, 0.05]
$\beta_{sppB.alleghaniensis}$	-0.08 [-0.12, -0.03]	-0.05 [-0.12, 0.02]	-0.08 [-0.23, 0.07]	-0.09 [-0.16, -0.01]
$\beta_{sppB.nigra}$	0.03 [-0.00, 0.05]	0.03 [-0.02, 0.07]	0.03 [-0.10, 0.16]	0.03 [-0.02, 0.08]
$\beta_{sppC.glabra}$	0.02 [-0.02, 0.06]	0.03 [-0.03, 0.08]	0.08 [-0.07, 0.23]	0.04 [-0.02, 0.09]
$\beta_{sppC.ovata}$	-0.00 [-0.05, 0.04]	-0.01 [-0.07, 0.05]	0.10 [-0.03, 0.24]	0.01 [-0.06, 0.08]
$\beta_{sppP.deltoides}$	0.00 [-0.02, 0.03]	0.02 [-0.03, 0.06]	0.07 [-0.09, 0.22]	0.02 [-0.02, 0.06]
$\beta_{sppQ.alba}$	0.02 [-0.06, 0.10]	0.03 [-0.06, 0.13]	0.00 [-0.18, 0.18]	0.05 [-0.07, 0.16]
$\beta_{sppQ.rubra}$	0.03 [-0.03, 0.09]	0.03 [-0.08, 0.14]	0.09 [-0.10, 0.28]	0.05 [-0.06, 0.15]
$\beta_{sppT.americana}$	-0.01 [-0.03, 0.01]	-0.02 [-0.05, 0.01]	0.15 [0.02, 0.27]	-0.01 [-0.04, 0.02]
$\alpha_{year2019}$	0.09 [-0.46, 0.64]	0.06 [-0.48, 0.63]	0.07 [-0.47, 0.64]	0.06 [-0.50, 0.62]
$\alpha_{year2020}$	-0.06 [-0.60, 0.49]	-0.07 [-0.61, 0.50]	-0.12 [-0.66, 0.45]	-0.07 [-0.63, 0.47]
$\alpha_{year2021}$	0.00 [-0.55, 0.56]	-0.00 [-0.55, 0.57]	0.01 [-0.54, 0.58]	-0.01 [-0.58, 0.54]
$\alpha_{year2022}$	-0.05 [-0.61, 0.50]	-0.07 [-0.63, 0.50]	-0.07 [-0.63, 0.49]	-0.08 [-0.65, 0.48]
$\alpha_{year2023}$	0.20 [-0.36, 0.76]	0.18 [-0.38, 0.75]	0.20 [-0.35, 0.78]	0.17 [-0.40, 0.72]
$\alpha_{year2024}$	0.08 [-0.47, 0.64]	0.07 [-0.47, 0.66]	0.07 [-0.48, 0.64]	0.08 [-0.49, 0.64]
$\alpha_{year2016}$	-0.21 [-0.77, 0.34]	-0.22 [-0.76, 0.35]	-0.25 [-0.80, 0.31]	-0.25 [-0.81, 0.29]
$\alpha_{year2017}$	-0.03 [-0.58, 0.52]	-0.06 [-0.59, 0.52]	0.00 [-0.54, 0.56]	-0.07 [-0.63, 0.48]
$\alpha_{year2018}$	0.04 [-0.52, 0.59]	0.02 [-0.52, 0.59]	-0.00 [-0.55, 0.56]	0.01 [-0.54, 0.56]
$\alpha_{sppA.rubrum}$	-0.05 [-3.40, 3.35]	-0.95 [-4.12, 2.17]	1.23 [-3.23, 5.60]	-1.08 [-5.62, 3.44]
$\alpha_{sppA.saccharum}$	-0.90 [-3.86, 2.10]	0.43 [-2.34, 3.17]	-0.57 [-3.59, 2.35]	-0.37 [-3.79, 3.02]
$\alpha_{sppAe.flava}$	-0.49 [-3.20, 2.19]	-0.46 [-3.09, 2.14]	0.93 [-2.69, 4.47]	-0.49 [-3.33, 2.38]
$\alpha_{sppB.alleghaniensis}$	2.32 [-0.68, 5.34]	0.61 [-2.29, 3.46]	1.17 [-2.34, 4.58]	3.25 [-0.47, 6.96]
$\alpha_{sppB.nigra}$	-0.68 [-3.54, 2.12]	-0.13 [-2.82, 2.58]	0.17 [-3.18, 3.40]	-0.50 [-3.65, 2.67]
$\alpha_{sppC.glabra}$	-0.89 [-3.89, 2.06]	-0.64 [-3.44, 2.12]	-1.17 [-4.65, 2.33]	-1.16 [-4.46, 2.12]
$\alpha_{sppC.ovata}$	-0.32 [-3.35, 2.74]	-0.07 [-2.88, 2.73]	-1.96 [-5.42, 1.40]	-0.35 [-4.01, 3.29]
$\alpha_{sppP.deltoides}$	-0.38 [-3.14, 2.40]	-0.39 [-3.09, 2.26]	-1.04 [-4.60, 2.52]	-0.49 [-3.54, 2.50]
$\alpha_{sppQ.alba}$	-0.85 [-4.70, 3.03]	-0.56 [-3.66, 2.52]	0.40 [-3.47, 4.32]	-1.44 [-6.32, 3.48]
$\alpha_{sppQ.rubra}$	-1.09 [-4.47, 2.23]	-0.43 [-3.80, 2.96]	-1.08 [-4.98, 2.91]	-1.44 [-6.04, 3.22]
$\alpha_{sppT.americana}$	0.30 [-2.41, 3.02]	0.59 [-1.98, 3.18]	-2.02 [-5.24, 1.14]	0.92 [-1.93, 3.81]

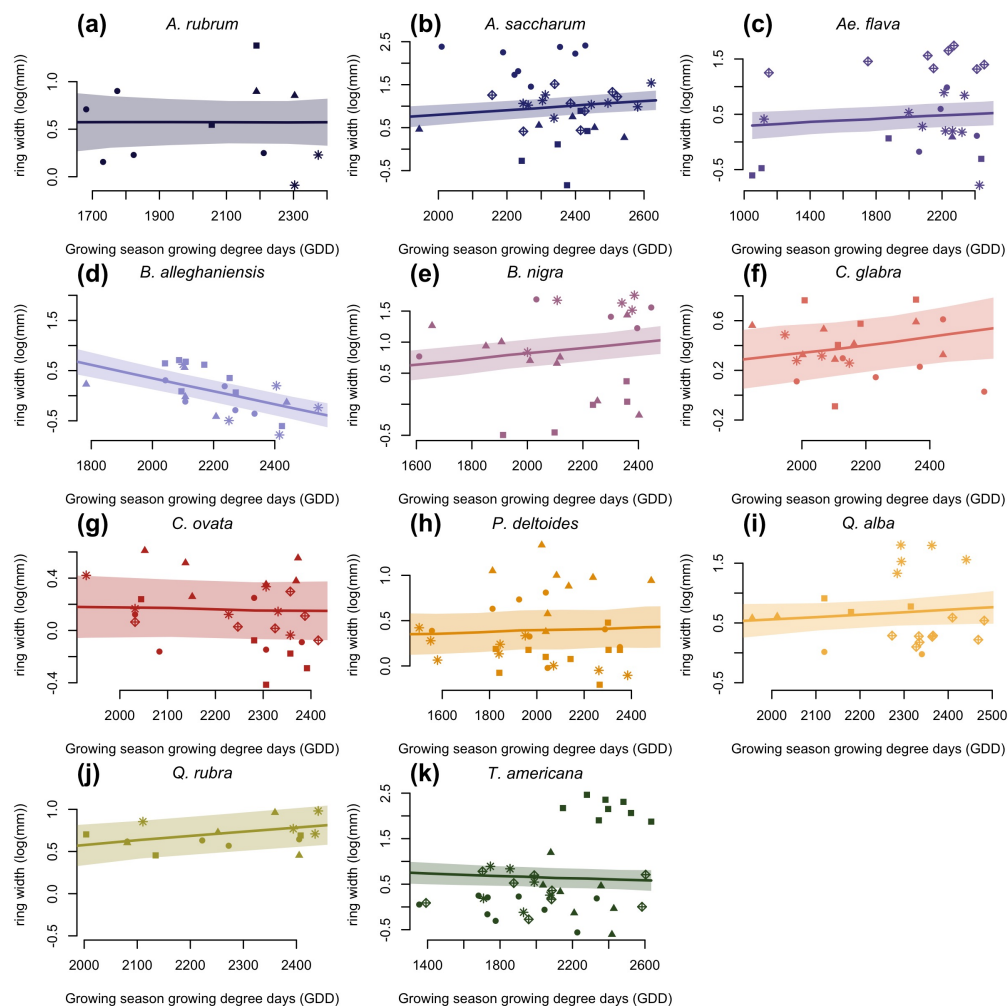


Figure S3: Ring width (on the log scale) trends in response to thermal season (GDD). The plots are the reconstructed species estimates posterior distribution with error. The lines represent the mean of the posterior distribution and the shaded area, the 50% UI. The dots are the empirical data shaped by treeids.

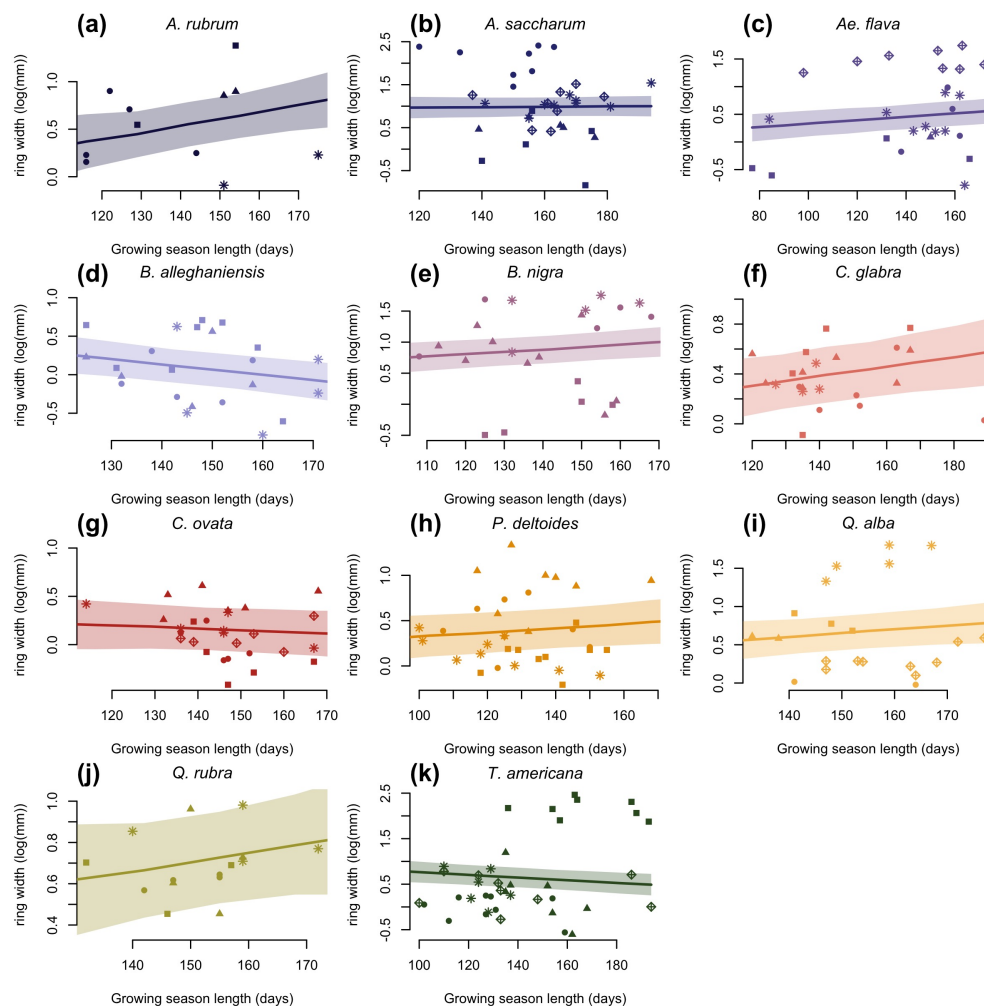


Figure S4: Ring width trends (on the log scale) in response to calendar season (GSL). The plots are the reconstructed species estimates posterior distribution with error. The lines represent the mean of the posterior distribution and the shaded area, the 50% UI. The dots are the empirical data shaped by treeids.

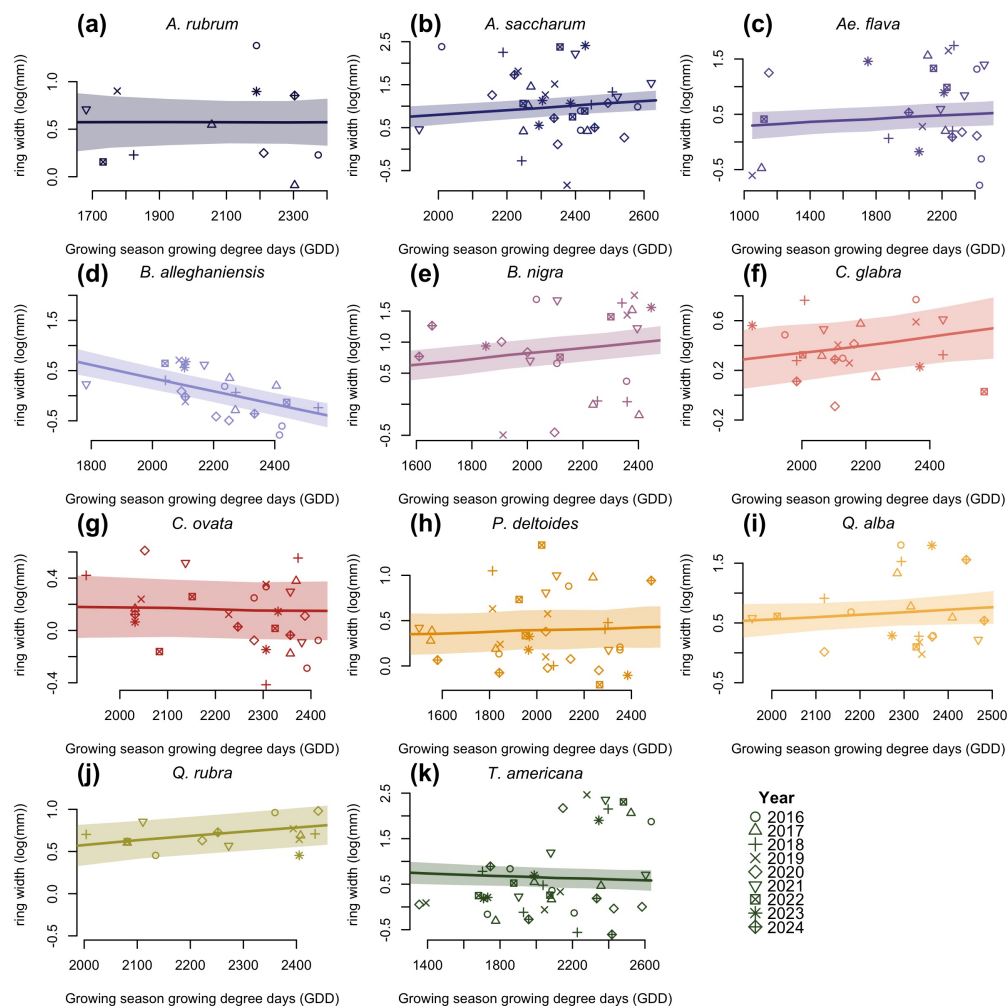


Figure S5: Ring width trends (on the log scale) in response to thermal season (GDD). The plots are the reconstructed species estimates posterior distribution with error. The lines represent the mean of the posterior distribution and the shaded area, the 50% UI. The dots are the empirical data shaped by year.

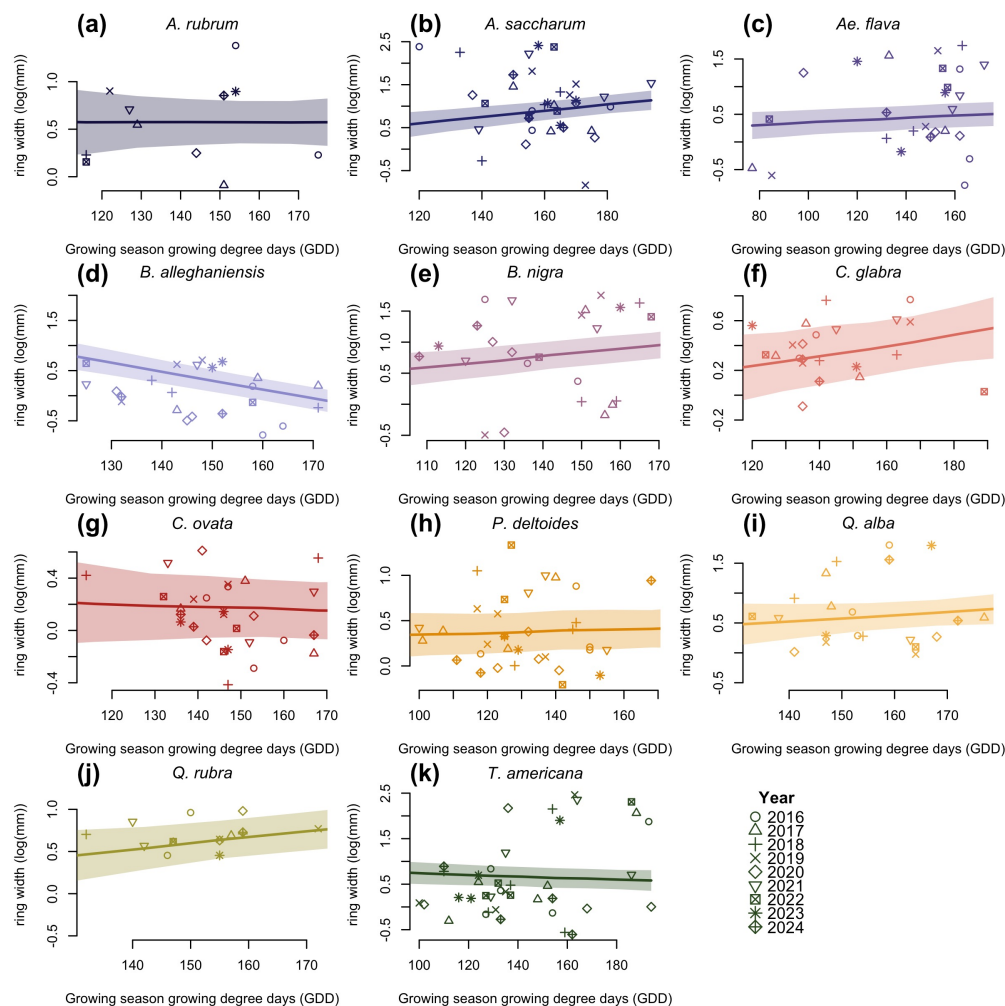


Figure S6: Ring width trends (on the log scale) in response to calendar season (GSL). The plots are the reconstructed species estimates posterior distribution with error. The lines represent the mean of the posterior distribution and the shaded area, the 50% UI. The dots are the empirical data shaped by year.

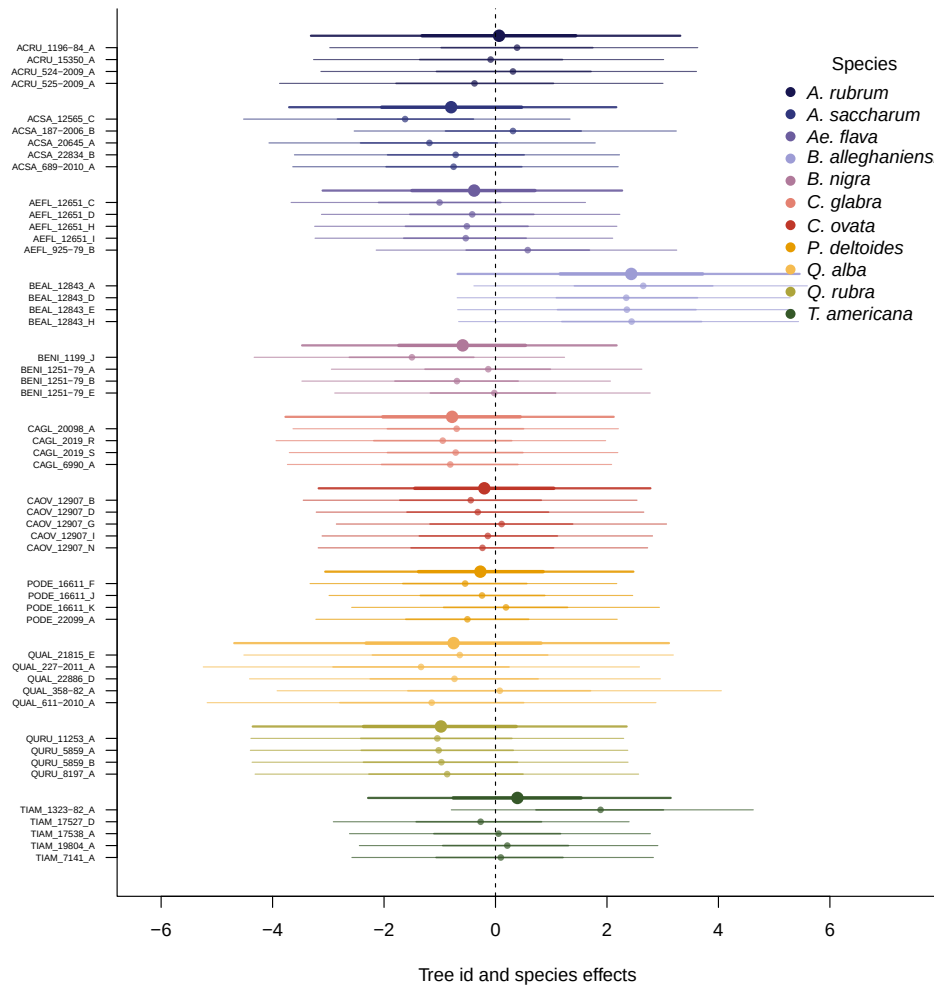
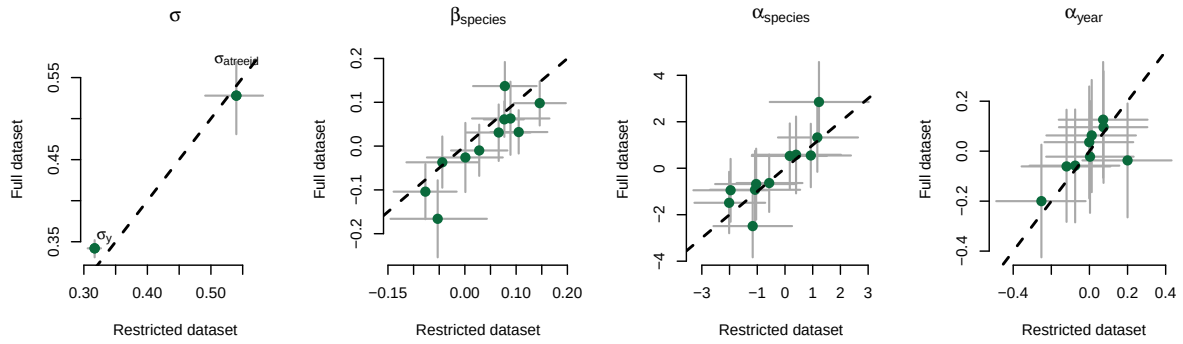


Figure S7: Ring width (on the log scale) deviations from the grand mean for each grouping factor of the GDD model. The different tree id values represent the combined intercepts of tree id, species and averaged year intercepts. The larger dots and lines represent the species intercepts which are the species level estimates averaged across all of their tree ids. The dots are the mean of the posterior distribution and the thick and thin lines, the 50 and 90% UI.

(a) Start of season (SOS)



(b) End of season (EOS)

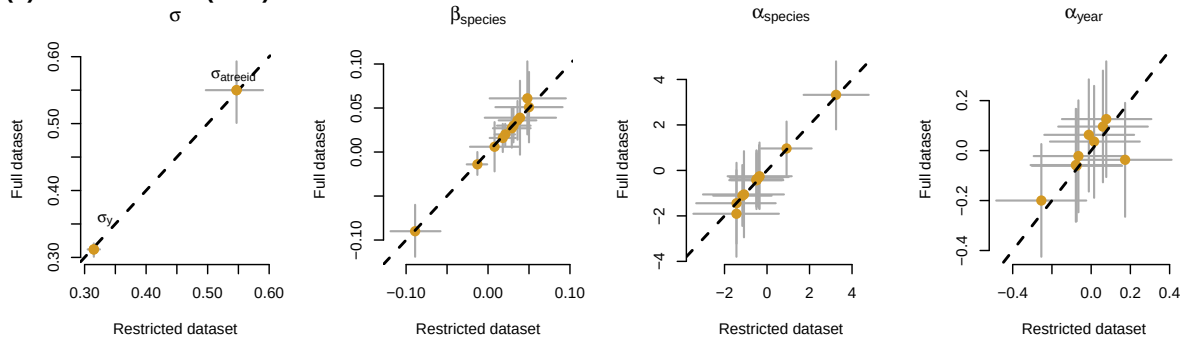


Figure S8: Full vs most restricted rows of data for the model with SOS (a) EOS (b) as the predictor. Each pannel represent the different parameters used to fit the models (see data analyses for more details).

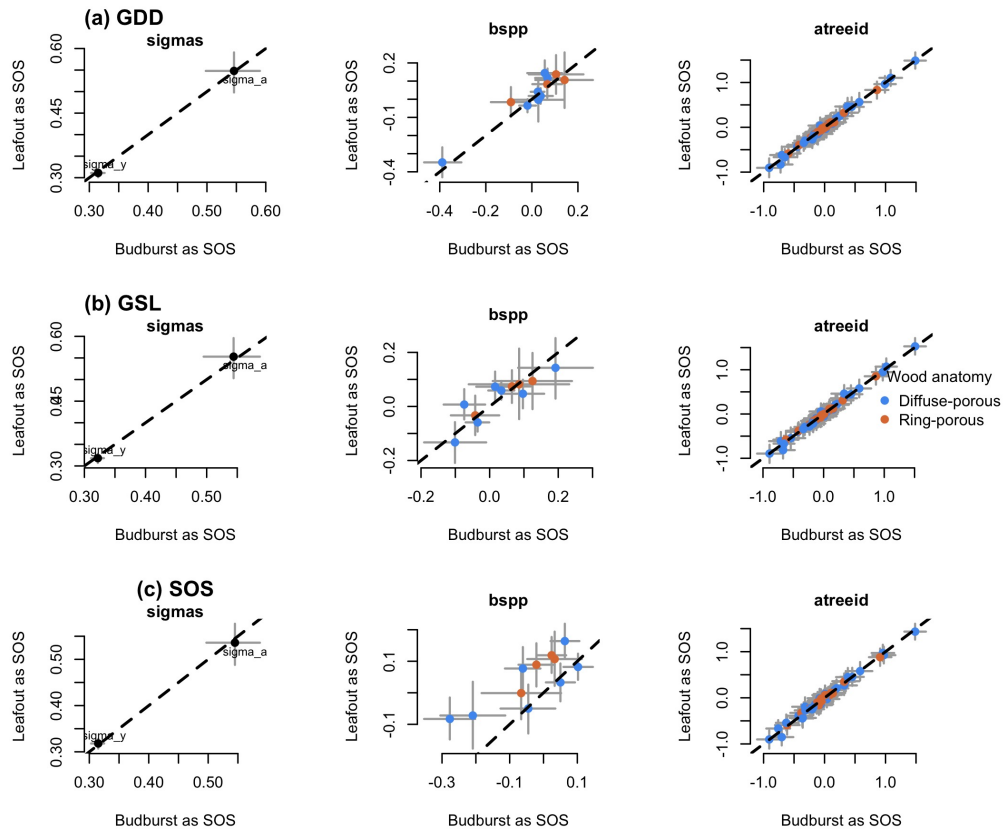


Figure S9: Parameter estimates when calculating the GDD (a), GSL (b) from leafout vs budburst, to leaf coloring. (c) shows the model estimates when taking leafout vs budburst as the SOS (see data analyses for more details).

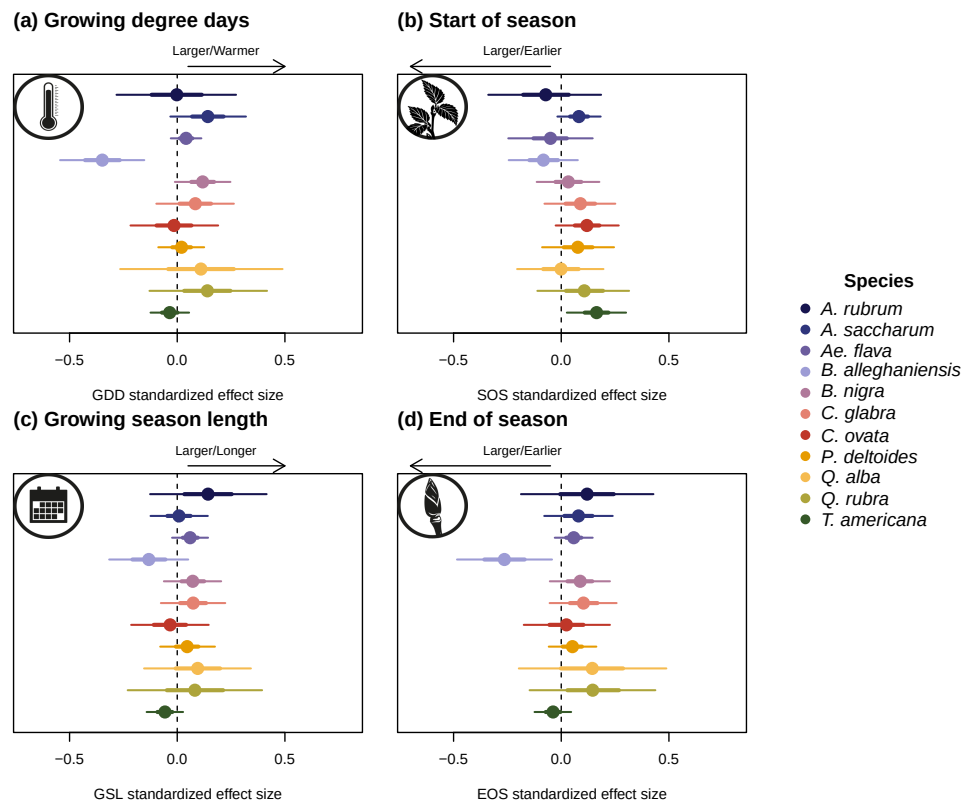


Figure S10: Effect size of each predictors across our eleven species. We depict ring width (on the log scale) change per scaled (z-scored; see data analyses section for more details) predictor for each species. We report estimates as mean and 90% uncertainty intervals (in square brackets) across our four different predictors (GDD: growing degree days, GSL: growing season length, SOS: start of season and EOS: end of season).

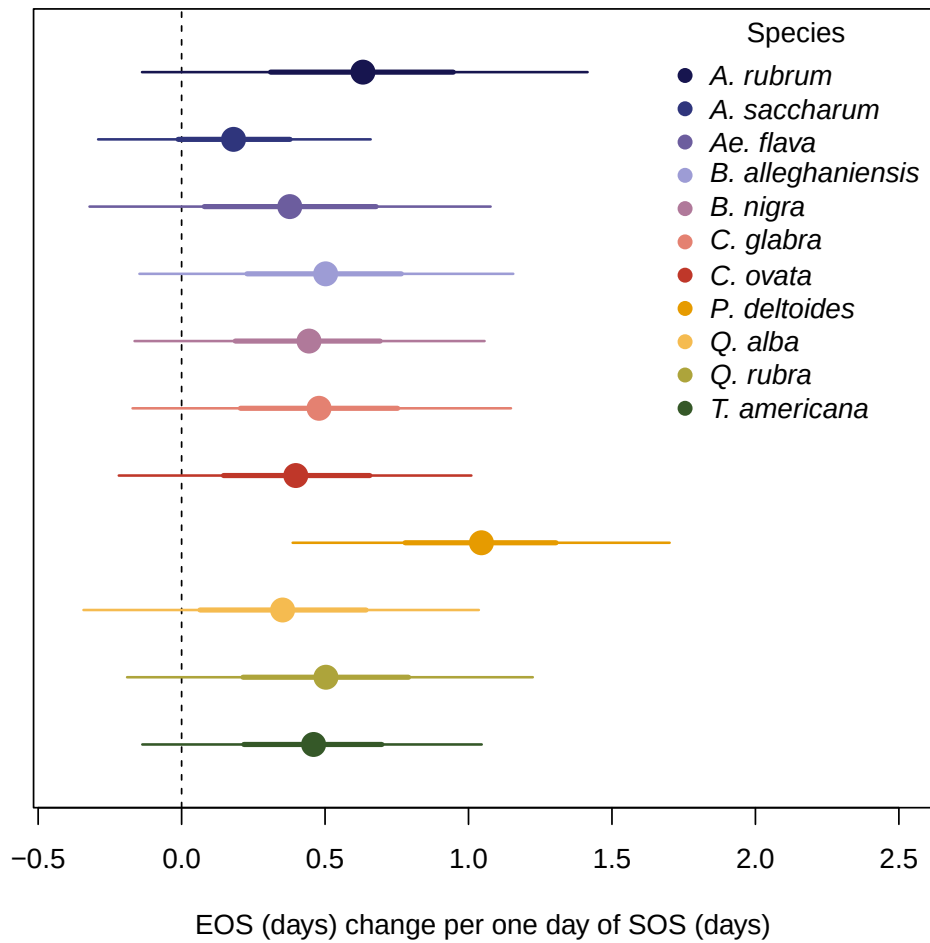


Figure S11: Effect of the start of season (SOS) on the end of season (EOS) across our eleven species. We report these slope estimates as mean and 90% uncertainty intervals.

Chapter 3

Concluding chapter

1. We used observational data to understand how longer and warmer seasons change growth for juvenile and mature trees
 - (a) Our findings are significant as they show that forest recruitment may not be harmed by climate change
 - (b) Mature trees store most of the forests carbon

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Appendix A

Supporting Materials

This would be any supporting material not central to the dissertation. For example:

- additional details of methodology and/or data;
- diagrams of specialized equipment developed.;
- copies of questionnaires and survey instruments.